
FMUSER FUQ-400 DTV Head-end Processor 6 Module Slots
ASI-IP-RF Interfaces UDP-RTP-RTSP
Encoding-Decoding-Multiplexing Descrambling Modulation for
CATV Systems



Product overview

The FMUSER FUQ-400 is a head-end processor with encoding, decoding, transcoding, multiplexing, descrambling, and modulation functions. It processes signals through six independent module slots for flexible configurations. Housed in a compact 1U chassis, it supports ASI and GE output interfaces and web-based management. Ideal for modern CATV systems requiring scalable and reliable solutions.

Key Features

Flexible Module Configuration: Supports up to six independent modules for customizable functions like encoding and modulation.

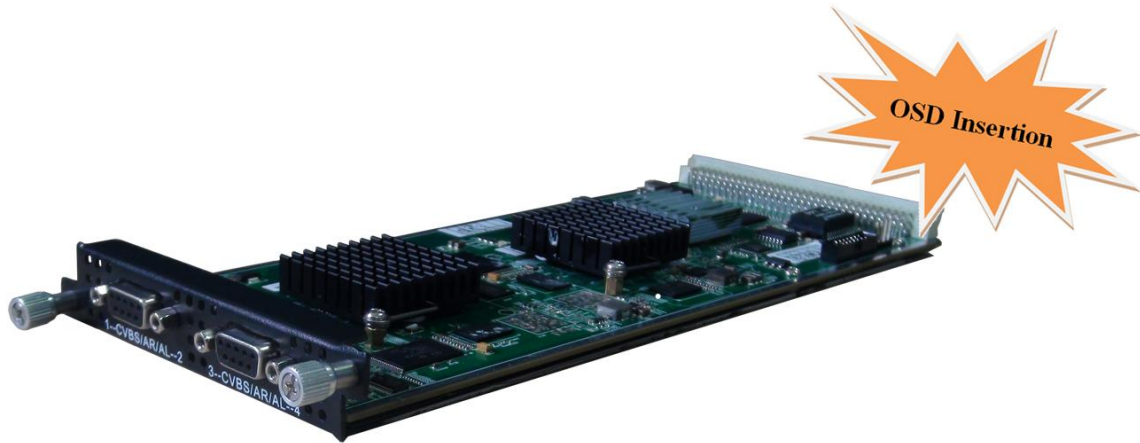
High-Capacity Output Interfaces: Includes ASI and GE ports, enabling 512 SPTS via GE1 and 8 MPTS via GE2 with UDP/RTP support.

Web-Based Management: Allows easy updates and control through a user-friendly interface.

Redundant Power Supply: Offers optional hot or non-hot plugging for enhanced reliability.

Module Specifications

4 CVBS/SDI Encoding Module (DX214B)



Module Specifications:

Input: 4 CVBS (DB9 to RCA) or 4 SDI (BNC)

Video Encoding:

Video format: MPEG-2, MPEG4 AVC/H.264

Image format: PAL, NTSC SD signal (Only for CVBS input)

Resolution:

*Input: 720*576 @50i*

*Output: 720*576/352*288/320*240/320*180/176*144/160*120/160*90@50Hz*

*Input: 720*480 @60i*

*Output: 720*480/352*288/320*240/320*180/176*144/160*120/160*90@60Hz*

Rate Control: CBR/VBR

GOP structure: IPPP, IBPBP, IBBPB, IBBBP

Video bitrate: 0.5~5Mbps

Audio Encoding:

Audio format: MPEG1 Audio Layer 2, LC-AAC, HE-AAC

Sampling rate: 48KHz

Bits per sample: 32-bit

Bit-rate: 48-384Kbps each channel

Support Logo, Caption, QR Code insertion

4 HDMI/SDI Encoding Module (DX224V)



Module Specifications:

Input: 4×SDI/HDMI (1.4) input, HDCP 1.4

Video Encoding:

Video format: HEVC/H.265 & MPEG 4 AVC/H.264

Resolution:

HDMI:

3840×2160_30P, 3840×2160_29.97P;

(Encoding 2 CHs per module for H.265, and encoding 1 CH for H.264)

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

1280×720_60P, 1280×720_59.94P, 1280×720_50P

(Encoding 4 CHs per module for H.264 and H.265)

SDI:

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

1280×720_60P, 1280×720_59.94P, 1280×720_50P

(Encoding 4 CHs per module for H.264 and H.265)

Input: 1920×1080_60i, 1920×1080_59.94i, 1920×1080_50i

Output: 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P

(Encoding 4 CHs per module for H.265, and encoding 2 CHs for H.264)

Chroma: 4:2:0 Bit rate: 0.5Mbps~20Mbps (each channel)

Rate Control: CBR/VBR GOP structure: IBBP, IPPP

Audio Encoding:

Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough

Sampling rate: 48KHz

Bit-rate (each channel):

48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC)

24 Kbps~128 Kbps (HE-AAC)

18 Kbps~56 Kbps (HE-AAC V2)

Audio Gain: 0~255

4 HDMI Encoding Module (DX224H/HV)



Module Specifications:

Input: 4×HDMI (1.4) input, HDCP 1.4

Video Encoding:

Video format: HEVC/H.265 & MPEG 4 AVC/H.264---DX224H

HEVC/H.265---DX224HV

Resolution:

1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P;

1280×720_60P, 1280×720_59.94P, 1280×720_50P

Input: 1920×1080_60i, 1920×1080_59.94i, 1920×1080_50i

Output: 1920×1080_60P, 1920×1080_59.94P, 1920×1080_50P

Chroma: 4:2:0 Bit rate (each channel): 0.5Mbps~20Mbps (H.265)

4 Mbps~20Mbps (H.264)

Rate Control: CBR/VBR GOP structure: IBBP, IPPP

Audio Encoding:

Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough

Sampling rate: 48KHz

Bit-rate (each channel):

48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC)

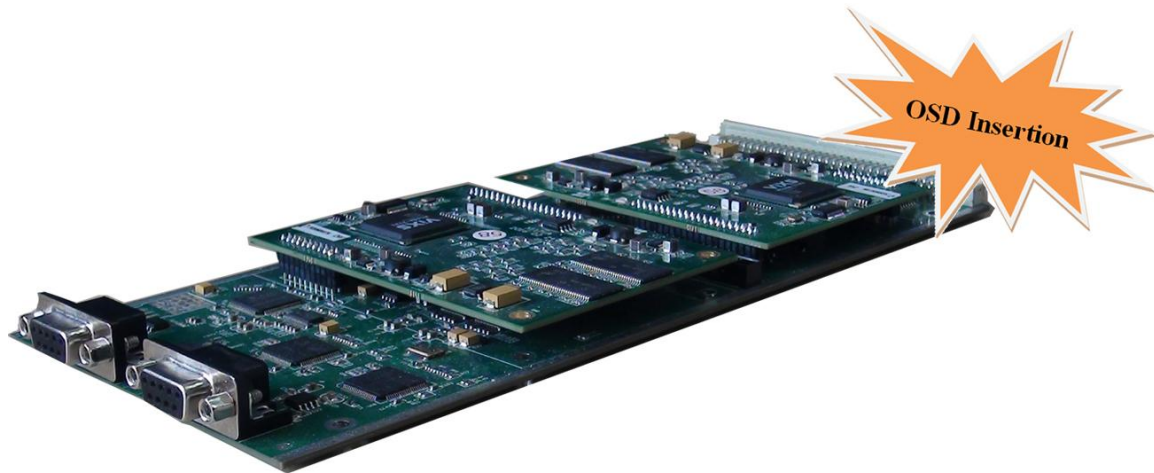
24 Kbps~128 Kbps (HE-AAC)

18 Kbps~56 Kbps (HE-AAC V2)

Audio Gain: 0~255

Support Logo, QR Code insertion--Optional as per order

4 CVBS Encoding Module (DX214/214A)



Module Specifications:

Input: 4 CVBS video, 4 Stereo Audio (DB9 to RCA)

Video Encoding:

Video format: MPEG-2 (4:2:0)

Image format: PAL, NTSC SD signal

Input resolution: 720×480_60i, 544×480_60i, 352×480_60i, 352×240_60i,
320×240_60i, 176×240_60i, 176×120_60i, 720×576_50i,
704×576_50i, 640×576_50i, 352×288_50i, 320×288_50i,
176×288_50i, 176×144_50i

GOP structure: IP, IBP, IBBP, IBBBP

Video bitrate: 0.5Mbps~8Mbps per channel

Support CC (closed caption)

Audio Encoding:

Audio format: MPEG-1 Layer 2, DD AC3 (2.0)

Sampling rate: 48KHz

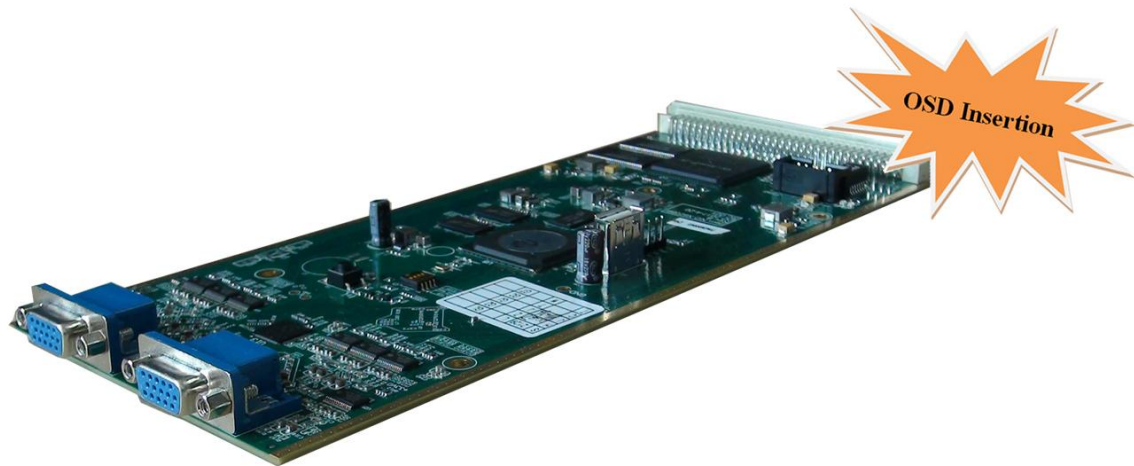
Resolution: 24-bit

Audio bitrate: 128/192/256/320/384kbps each channel

Support Logo, Caption, QR Code insertion (for DX214A only)

(Language Supported: ■■, English, ■■■■, for more languages please consult us...)

8 CVBS Encoding Module (DX218S)



Module Specifications:

Input: 8 CVBS video, 8 Stereo Audio (DB15 to RCA)

Video Encoding:

Video format: MPEG4 AVC/H.264

Image format: PAL, NTSC SD signal

Resolution: 720×576i, 720×480i

Rate Control: CBR/VBR

GOP structure: IPP

Video bitrate: 1~7Mbps each channel

Audio Encoding:

Audio format: MPEG-1 Layer 2

Sampling rate: 48KHz

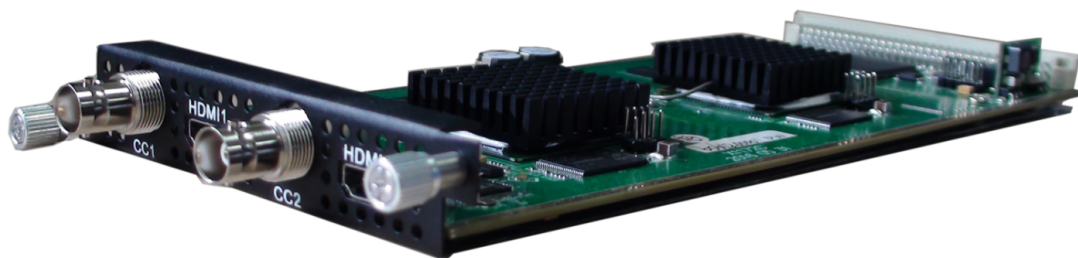
Resolution: 24-bit

Bit-rate: 64/128/192/224/256/320/384Kbps each channel

Support Logo, Caption, QR Code insertion

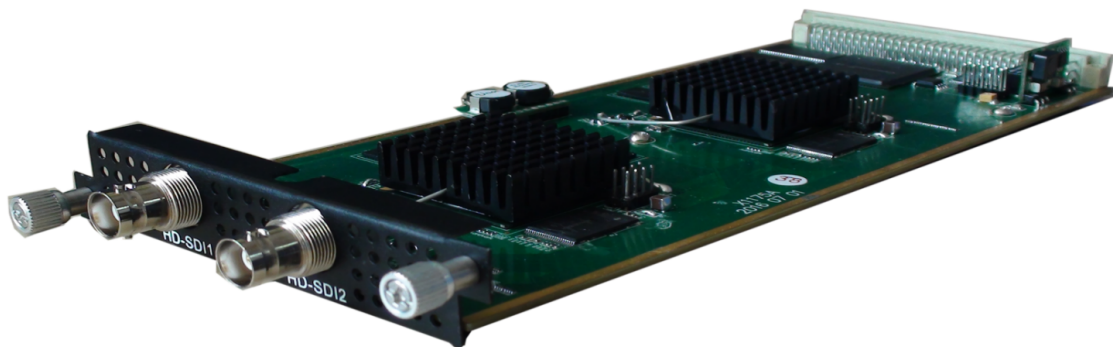
(Language Supported: ■■■, English, ■■■■■, for more languages please consult us...)

2 HDMI Encoding/Transcoding Module (DX202A)



Module Specifications:

*Input: 2*HDMI, 2*BNC for CC (Closed Caption) input and stream connector*

*Video Encoding:**Video format: MPEG2 & MPEG4 AVC/H.264**Input resolution:**1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i**Rate control mode: CBR/VBR**Aspect ratio: 16:9, 4:3**Video bitrate: 0.8~19Mbps for MPEG-2 /H.264 encoding**Support CC (closed caption)**Audio Encoding:**Audio format: MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC,**Dolby Digital AC3 (2.0) encoding (Optional); AC3 (2.0/5.1) passthrough**Sampling rate: 48KHz**Audio bitrate: 64Kbps-320kbps each channel**Video Transcoding:**2*MPEG2 HD → 2*MPEG2/H.264 HD; 2*MPEG2 HD → 2*MPEG2/H.264 SD;**2* H.264 HD → 2*MPEG2/H.264 HD; 2* H.264 HD → 2*MPEG2/H.264 SD;**4 *MPEG2 SD → 4 *MPEG2/H.264 SD; 4* H.264 SD → 4 *MPEG2/H.264 SD**Audio Transcoding:**MPEG-1 Layer 2, AC3 (Optional) and AAC any-to-any**2 HD-SDI Encoding/Transcoding Module (DX202A-D)**Module Specifications:**Input: 2*HD-SDI and stream connector**Video Encoding:**Video format: MPEG2 & MPEG4 AVC/H.264**Input resolution:*

1920*1080_60P, 1920*1080_50P, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i

Rate control mode: CBR/VBR

Aspect ratio: 16:9, 4:3

Video bitrate: 0.8~19Mbps for MPEG-2 /H.264 encoding;

Support CC (closed caption)

Audio Encoding:

Audio format:

MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC,

Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough

Sampling rate: 48KHz

Audio bitrate: 64Kbps-320kbps each channel

Video Tanscoding:

2*MPEG2 HD → 2*MPEG2/H.264 HD; 2*MPEG2 HD → 2*MPEG2/H.264 SD;

2* H.264 HD → 2*MPEG2/H.264 HD; 2* H.264 HD → 2*MPEG2/H.264 SD;

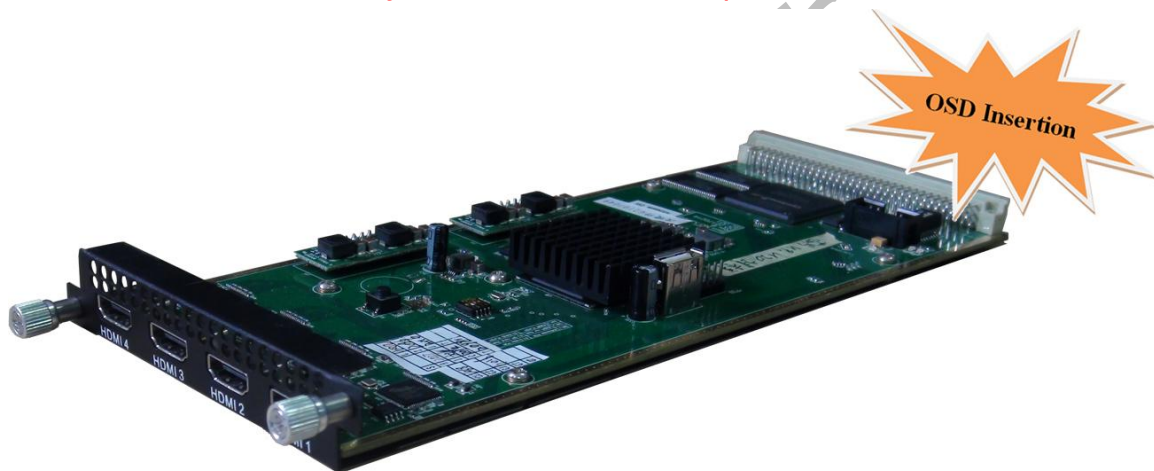
4 *MPEG2 SD → 4 *MPEG2/H.264 SD; 4* H.264 SD → 4 *MPEG2/H.264 SD

Audio Tanscoding:

MPEG-1 Layer 2, AC3 (Optional) and AAC any-to-any

4 HDMI Encoding Module (DX224HS)

No Stock Available Currently. Please contact sales person for details.



Module Specifications:

Input: 4*HDMI (1.4) input, HDCP 1.4

Video Encoding:

Video format: HEVC/ H.265, MPEG4 AVC/H.264

Input Resolution	Available Output Resolution
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1920x1080P@60/59.94	1920x1080/1280x720/720*576/720*480P@60/50/30/25/24
1920x1080I@60/59.94	1920x1080/1280x720/720*576/720*480P@30/25/24
1920x1080P@50	1920x1080/1280x720/720*576/720*480P@50/30/25/24
1920x1080I@50	1920x1080/1280x720/720*576/720*480P@25/24
1280x720P@60/59.94	1280x720/720*576/720*480P@60/50/30/25/24
1280x720P@50	1280x720/720*576/720*480P@50/30/25/24
720*576I@50	720*576/720*480P@25/24
720*480I@60	720*480P@30/25/24

Chroma: 4:2:0

GOP structure: IP, IBBP, IBBBP

Video Bit-rate: 1Mbps~15Mbps per channel

Rate Control: CBR/VBR

Audio Encoding:

Audio format: MPEG-1 Layer 2, LC-AAC, HE-AAC, HE-AAC V2, AC3 Passthrough

Sampling rate: 48 KHz

Resolution: 24-bit

Audio Bit-rate (per channel):

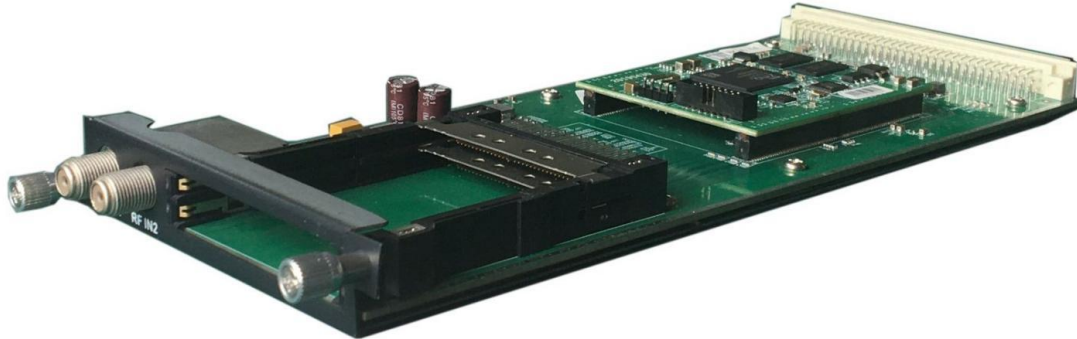
48Kbps~384Kbps (MPEG-1 Layer 2 & LC-AAC)

24 Kbps~128 Kbps (HE-AAC)

18 Kbps~56 Kbps (HE-AAC V2)

Others: Support Logo, Caption, QR Code insertion

2 Tuner Descrambling Module (DX902A)



Module Specifications:

Stream in: 2 Tuner input, F Type

DVB-CI: 2 Independent common interface slots

Standard: DVB-S/S2/S2X

Tuner Section:

DVB-S Input Frequency: 950-2150MHz

Symbol Rate: QPSK 0.5~45 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2 Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 32APSK 0.5~40 Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

32APSK: 3/4, 4/5, 5/6, 8/9

DVB-S2X *Input Frequency: 950-2150MHz*

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 8APSK/32APSK: 0.5~40Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L, 28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15,

7/9

Signal Strength: -65~ -25dBm

Support Diseqc function

Multiplexing:

Maximum PID Remapping: 256 output pids

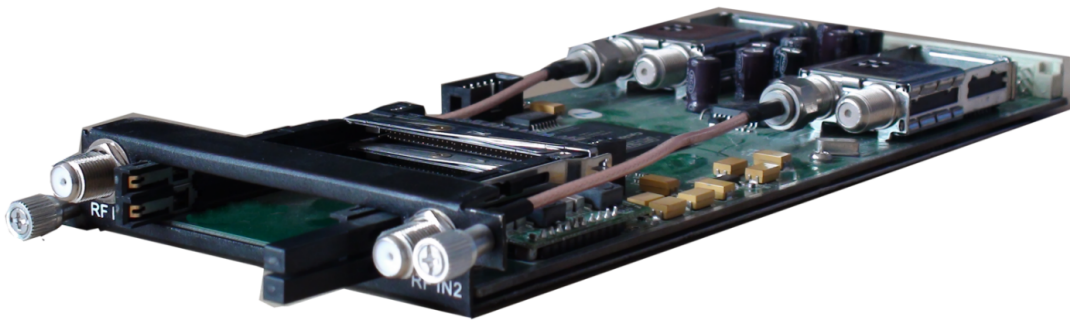
Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; up to 120Mbps (Optional as required)

2 Tuner Descrambling Module (DX912)



Module Specifications:

Stream in: 2 Tuner input, F Type

DVB-CI: 2 Independent common interface slots

Standard: DVB-C (J.83 A/C)/J.83B/ DVB-T/ISDB-T switchable

Standard: DVB-C (J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP Index: 0~255 (Optional)

Standard: ISDB-T

Input Frequency: 60-890MHz

Signal Strength: -65~ -25dBm

Multiplexing:

Maximum PID Remapping: 256 output pids

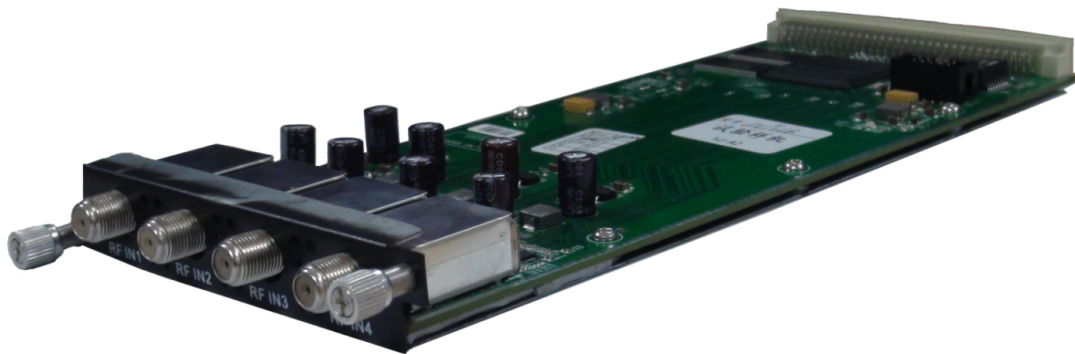
Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Descrambling:

CAM/CI Quantity: 2

BISS Mode: Mode 1, Mode E; up to 120Mbps (Optional as required)

4 FTA Tuner Module (DX904B)



Module Specifications:

Stream in: 4 Tuner input, F Type

Stream out: 1 MPTS out over UDP, unicast/multicast

Standard: DVB-S/S2/S2X

Tuner Section

DVB-S Input Frequency: 950-2150MHz

Symbol Rate: QPSK 0.5~45 Msps

FEC Demodulation: 1/2, 2/3, 3/4, 5/6, 7/8

DVB-S2 Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 32APSK 0.5~40 Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10

32APSK: 3/4, 4/5, 5/6, 8/9

DVB-S2X Input Frequency: 950-2150MHz

Symbol rate: QPSK/8PSK/16APSK 0.5~45 Msps; 8APSK/32APSK: 0.5~40Msps

FEC Demodulation: QPSK: 1/2, 3/5, 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 13/45, 9/20, 11/20

8PSK: 3/5, 2/3, 3/4, 5/6, 8/9, 9/10, 23/36, 25/36, 13/18

8APSK: 5/9-L, 26/45-L

16APSK: 2/3, 3/4, 4/5, 5/6, 8/9, 9/10, 1/2-L, 8/15-L, 5/9-L, 26/45, 3/5, 3/5-L,
28/45, 23/36, 2/3-L, 25/36, 13/18, 7/9, 77/90

32APSK: 3/4, 4/5, 5/6, 8/9, 2/3-L, 32/45, 11/15, 7/9

Signal Strength: -65 ~ -25dBm

Support Diseqc function

Multiplexing:

Max number of managing PID: 256

Function: PID remapping (automatically/ manually), Accurate PCR adjust, PID pass-through

4 FTA Tuner Module (DX924)



Module Specifications:

Stream in: 4 Tuner input, F Type

Standard: DX924: DVB-C(J.83 A/C)/J.83B/ DVB-T/ISDB-T switchable

Standard: DVB-C(J.83 A/C); J.83B

Input Frequency: 60MHz~890MHz

Symbol rate: 1000~9000Ksps

Constellation: 16/32/64/128/256 QAM; 64/256 QAM for J.83B

Standard: DVB-T

Frequency In: 60MHz~890MHz

Bandwidth: 5/6/7/8M bandwidth

PLP Index: 0~255(optional)

Standard: ISDB-T

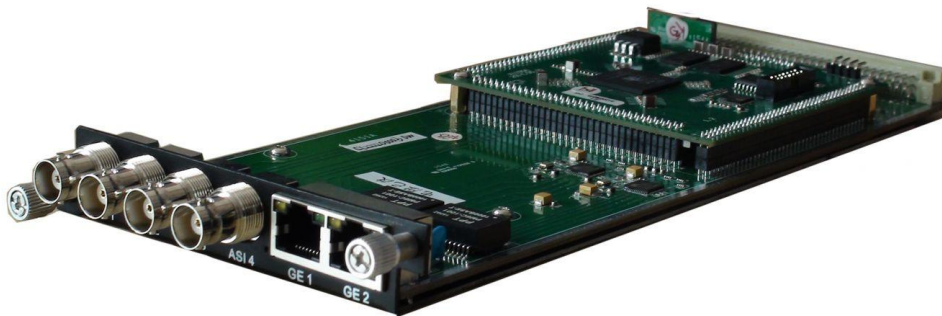
Input Frequency: 60-890MHz

Multiplexing:

Max number of managing PID: 256

Function: PID remapping (automatically/ manually), Accurate PCR adjust, PID pass-through

4 ASI/IP Multiplexing Module (DX504)



Module Specifications:

ASI inputs/outputs: 4 ASI bi-direction, BNC 75Ω

IP inputs/outputs: 2 Ethernet Port(100/1000M)

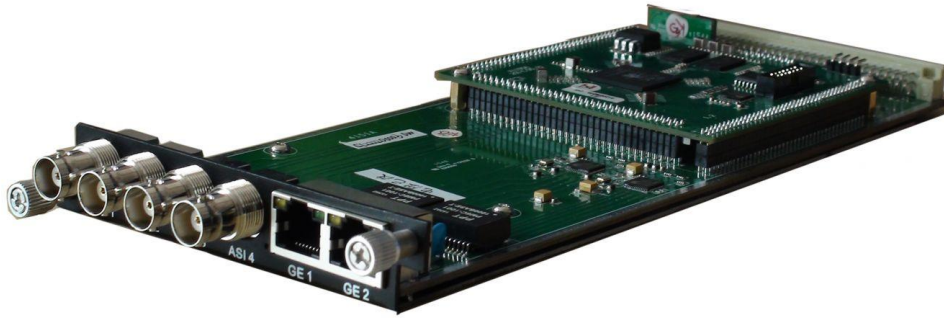
Stream connector input

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

Stream In: maximum 4 ASI input, 256×2 IP input

Stream Out: maximum 4 ASI output, 4 IP output

4/8 CH EAS Multiplexing Module (DX504E/DX508E)



DX504E/DX508E Module Specifications:

ASI input: ASI1 input (SPTS), BNC 75Ω

IP input: 256 IP input thru 1 GE1 (100/1000M)

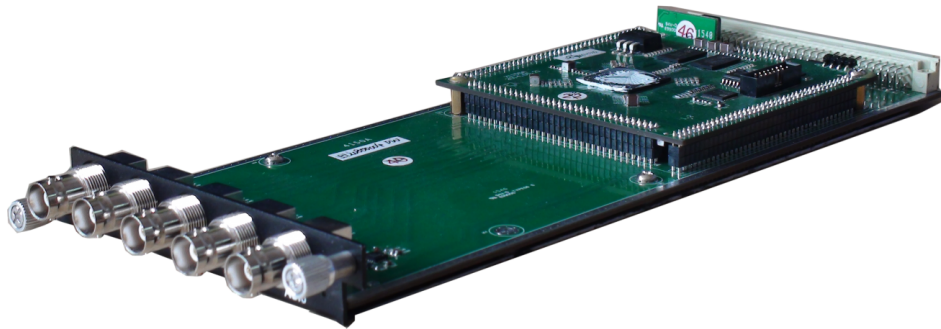
EAS Source: ASI1 or IP (the 256th IP) (ASI&IP should be SPTS, both can't mux, source Bitrate≤10Mbps)

Re-multiplexing: PID remapping, PCR correction (only for IP), generate PSI/ SI table automatically

Stream Out: 4 IP output thru GE1, maximum 16 programs each channel--DX504E

Stream Out: 8 IP output thru GE1, maximum 8 programs each channel--DX508E

5 ASI Multiplexing Module (DX505)



Module Specifications:

ASI inputs/outputs: 5 ASI bi-direction, BNC 75Ω

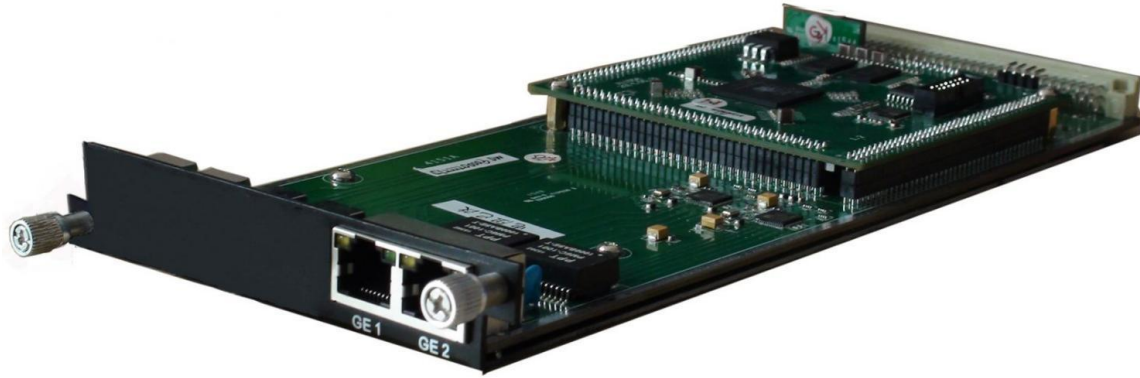
Stream connector input

Stream in: maximum 5 ASI input

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

Stream out: maximum 5 ASI output

IP Multiplexing Module (DX506)



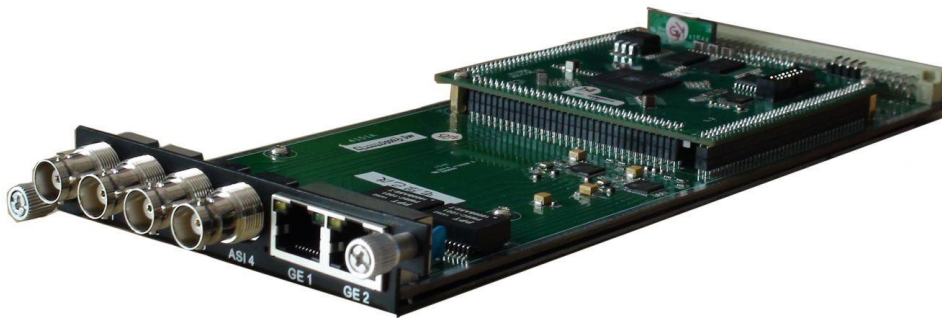
Module Specifications:

IP input: 512 SPTS or MPTS input over UDP, RTP, Unicast and Multicast thru GE1 Ethernet Port(100/1000M)

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

Stream Out: 512 SPTS output over UDP, RTP, Unicast and Multicast thru GE2 Ethernet Port(100/1000M)

ASI/IP Mux-Scrambler Module (DX514)



Module Specifications:

ASI inputs/outputs: 4 ASI bi-direction, BNC 75Ω

IP inputs/outputs: 2 Ethernet Port (100/1000M)

Stream connector input

Re-multiplexing: PID remapping, PCR correction, generate PSI/ SI table automatically

Stream In: maximum 4 ASI input, IP input over UDP/RTP/

Stream Out: maximum 4 ASI output, 4 IP (MPTS) output over UDP/RTP/RTSP

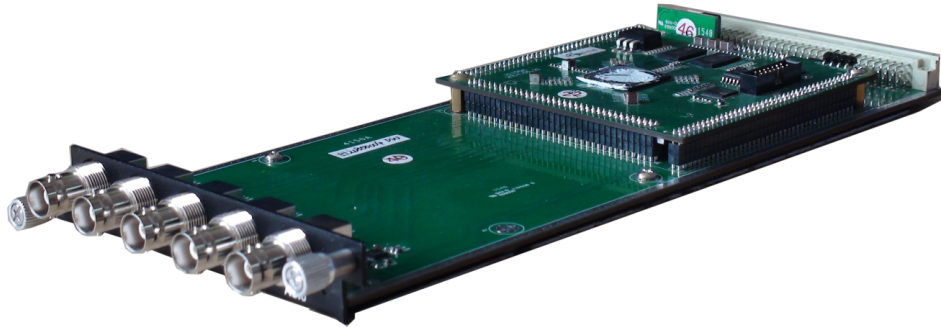
Scrambling:

Maximum simulcrypt CA: 4

Standard: ETR289, ETSI 101 197, ETSI 103 197

Connection: Local/remote connection

ASI TS Switch Module (DX515)



Module Specifications:

Input: 3 ASI

Output: 2 ASI mirror out

TS packet: 188/204

Switch Mode: Manually/Auto

Support monitor program status (bitrate, continuous counting, synchronization status)

Support channel mode and program mode switching

Support to switch back to the main stream manually or automatically once the main stream recovers

16/32 QAM Modulating Module (DX316/DX332)



Module Specifications:

Data input: 512x2 IP input over UDP/RTP, 2 GE Ports (RJ45/SFP), Stream connector

Data output: 16 or 32 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP)

Trans Rate: Max 840Mbps/GE Port

RF output (F type): 16/32 channels of multiplexing, scrambling and modulation.

Multiplexing:

Maximum PID Remapping: 180 output pids per channel for DX316, 256 output pids per channel for DX332

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/SI table automatically

Scrambling:

Maximum simulcrypt CA: 4

Standard: ETR289, ETSI 101 197, ETSI 103 197

Connection: Local/remote connection

Modulation:

Standard: EN300 429/ITU-T J.83A/B (DVB-C)

MER: $\geq 40\text{dB}$

RF frequency: 50~960MHz, 1KHz step

RF output level: $-20\sim+10\text{dBm}$ ($87\sim117\text{ dB}\mu\text{V}$), 0.1dB step for all carriers

Symbol Rate: 5.0Msps~7.0Msps, 1ksps stepping

Constellation: 16/32/64/128/256QAM

DX316 Output: 16 non-adjacent carrier outputs within 192M bandwidth

DX332 Output: 32 non-adjacent carrier outputs within 384M bandwidth

48 QAM Modulating Module (DX348D)



Module Specifications:

Data input: 1024 IP input over UDP/RTP, unicast/multicast, 2 GE Ports (RJ45/SFP), 512 IP input through stream connector

Data output: 48 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP), GE1 for IP #1-#24, GE2 for IP #25-#48

Trans Rate: Max 840Mbps/GE Port

RF output: 2 F type output ports for 48 non-adjacent carriers, 75 Ω

Carrier 1~24 out thru RF1, 25~48 thru RF2, each RF supports 384M bandwidth

Multiplexing:

Maximum PID Remapping: 256 output PIDs per channel

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/SI table automatically

*Scrambling:**Maximum simulcrypt CA: 6**Standard: ETR289, ETSI 101 197, ETSI 103 197**Connection: Local/remote connection**Modulation:**Standard: EN300 429/ITU-T J.83A/B/C MER: ≥ 40 dB**RF frequency: 50~960MHz, 1KHz step**RF output level: -20~+10dBm (87~117 dB μ V), 0.1dB step for all carriers**Symbol Rate: 3600~7000Ksps, 1ksps stepping**5057Ksps (J.83B, 64QAM), 5361Ksps (J.83B, 256QAM)**Constellation/Bandwidth: J.83A: 16/32/64/128/256QAM, 8M**J.83B/C: 64/256QAM, 6M**8 DVB-T/ATSC Modulating Module (DX308T/DX308AT)**Module Specifications:**Data input: Stream connector**256 IP input over UDP/RTP, 2GE Ports (RJ45/SFP) — DX308T**256 IP input over UDP/RTP, 2GE Ports (RJ45/SFP) — DX308AT**Data output: 8 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP)**Trans Rate: Max 840Mbps/GE Port**RF Output (F type): 8 non-adjacent carrier outputs within 192M bandwidth**Multiplexing:**Channel Number: 8 multiplexing channels**Maximum PID Remapping: 180 output pids per channel**Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically**Modulation: DX308T (8*DVB-T)*

Standard: ETSI EN300 744 MER: $\geq 40\text{dB}$

RF Frequency: 50~960MHz, 1KHz step

Constellation: QPSK/16QAM/64QAM

Bandwidth: 6/7/8 MHz

Trans mode: 2K/4K/8K FEC: 1/2, 2/3, 3/4, 5/6, 7/8

RF Output Level: -20~+10dBm (for all carriers), 0.5dB step

Modulation: DX308AT (8*ATSC)

Standard: ATSC A/53 MER: $\geq 40\text{dB}$ RF Frequency: 50~960MHz, 1KHz step

Constellation: 8VSB Bandwidth: 6MHz FEC: RS(208 188)+Trellis

RF Output Level: -20~+10dBm (for all carriers), 0.5dB step

6 ISDB-Tb Modulating Module (DX306I)



Module Specifications:

Data input: 32x6 IP input over UDP/RTP, 2 GE Ports (RJ45/SFP) and stream connector

Data output: 6 IP output over UDP/RTP/RTSP, unicast/multicast, 2 GE Ports (RJ45/SFP)

Trans Rate: Max 840Mbps/GE Port

RF output (F type): 6 channels of multiplexing and modulation.

Multiplexing:

Maximum PID Remapping: 180 output pids per channel

Function: PID remapping (automatically or manually), Accurate PCR adjusting, generate PSI/ SI table automatically

Modulation:

Standard: ARIB STD-B31

Bandwidth: 6M

Constellation: QPSK, 16QAM, 64QAM

Guard Interval: 1/32, 1/16, 1/8, 1/4

Transmission Mode: 2K, 4K, 8K

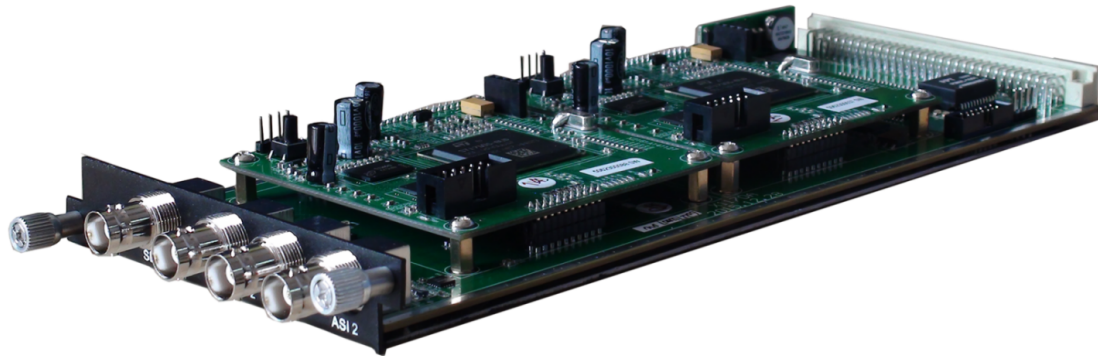
Code rate: 1/2, 2/3, 3/4, 5/6, 7/8

MER: $\geq 40\text{dB}$

RF frequency: 50~960MHz, 1KHz step

RF output level: -20dBm~+10dBm (87~117dBμV), 0.1dB stepping

2 HD-SDI Decoding Module (DX702)



Module Specifications:

ASI input/output: 2 ASI bi-direction, BNC 75Ω

Stream Connector input

Decoding:

Video/Audio Out: 2 HD/SD SDI output

Video Format: MPEG-2, MPEG-4 AVC/H.264

Resolution: 480i, 480p, 576i, 576p, 720p@50/59.94/60, 1080i@50/59.94/60

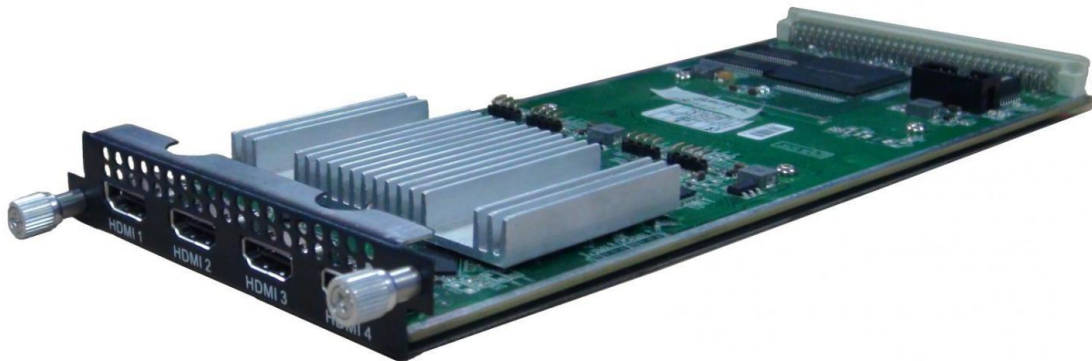
Chroma: 4:2:0

Audio Format: MPEG1 Layer2, LC-AAC, HE-AAC, AC3 (2.0/5.1), AC3 Passthrough,

Support Dual Audio Out

Support CC/Subtitle

4 HDMI Decoding Module (DX714)



Module Specifications:

Input: 1 IP (MPTS/SPTS) input over UDP thru stream connector, Unicast/Multicast

Output: 4 HDMI output

Decoding:

Video/Audio Out: 4 HDMI output with 1 channel stereo audio embedded in each port

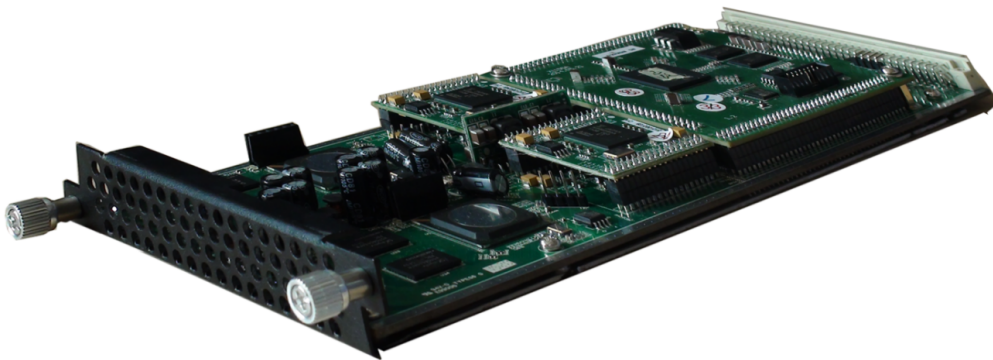
Video Format: MPEG-2, MPEG-4 AVC/H.264, HEVC/H.265, AVS, AVS+

Audio Format: MPEG 1 Layer 2, LC-AAC, HE-AAC, AC3 (2.0)

Resolution: 480i, 480p, 576i, 576p, 1280×720_50p, 1280×720_60p, 1920×1080_50i, 1920×1080_50p, 1920×1080_60i, 1920×1080_60p

Support manually upscale/downscale resolution

2 IP Transcoding Module (DX202)

*Module Specifications:*

Input: Stream connector

Resolution: 480i, 576i, 720P@50, 720P@60, 1080i@50, 1080i@60, 1080P@50, 1080P@60

Video Transcoding:

*2*MPEG-2/ H.264/ AVS/AVS+ HD/SD → 2* H.264 HD/SD*

Video Bit-rate: 1~19.5Mbps each channel

Rate Mode: CBR/VBR

GOP Struct: IBBP, IPPP, IBP

Audio Transcoding:

MPEG-1 Layer II, LC/HE-AAC, AC3, DRA → MPEG-1 Layer II, LC/HE-AAC

Audio bitrate: 64/96/128/192/256/320/384Kbps

Audio Gain Control: 0-100

Equipment Specifications:

Base Unit Parameters

Dimension(W×L×H): 482mm×410mm×44mm

Environment: 0~45■(work); -20~80■(Storage■

Power requirements AC 110V± 10%, 50/60Hz, AC 220 ± 10%, 50/60Hz

Parameters Comparison:

	DHP400	DHP400A
IP Data Transport	One-way Transport: GE1/GE2 support output maximum 8MPTS & 512SPTS	Bi-directional Transport: GE1/GE2 support IP data input and output
TS Processing Number	Support 1 TS (MPTS/SPTS) input from each module, and maximum bitrate is 350Mbps	Support maximum 512 TS (MPTS/SPTS) input from each module, and maximum bitrate is 350Mbps
Multiplexing	Support multiplexing function: It can mux TSs from different modules to one TS and output through one module or GE1/GE2 port	Doesn't support multiplexing function: It can combine TSs from different modules and output these TSs through one module or GE1/GE2 port
Output Per Module	1 MPTS after multiplexing	1 or multiple MPTS/SPTS

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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